# BOKU Power BI Migration

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# Problem Framing

## LEGACY SYSTEM

- Oracle Warehouse Builder (ETL)
- SAP BusinessObjects (Reporting)
- 20+ years of accumulated complexity

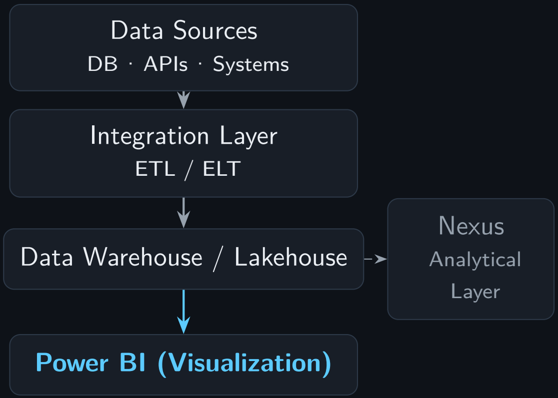
## CHALLENGES

- Fragmented ETL and reporting layers
- Limited scalability and flexibility
- No path to analytics or AI integration

Replace with a modern, scalable, unified data platform

# Target Architecture

## ARCHITECTURE LAYERS




## KEY PRINCIPLE

- Power BI = **Visualization Layer only**
- Logic and transformations **live upstream**
- Clear separation: Ingestion · Storage · Viz

# Migration Strategy

### PHASE 1 — INVENTORY

- Catalogue reports and ETL jobs
- Classify: Critical / Replaceable / Obsolete
- Identify data owners

### PHASE 2 — PARALLEL BUILD

- New system runs alongside legacy
- Define clear integration points
- Validate via pilot reports
- **No Big Bang**

### PHASE 3 — CONTROLLED CUTOVER

- Stepwise migration (per report cluster)
- Rollback always available
- Legacy stays active (transition phase)

# Risks & Mitigation

### TECHNICAL

- Hidden ETL dependencies
- Data inconsistencies
- KPI definition mismatch
- Performance deviations

### ORGANIZATIONAL

- User resistance (20+ years habits)
- Misaligned expectations
- Training gap

### MITIGATION

- Automated validation (old vs. new)
- Early key-user involvement
- Versioned datasets
- Remove unused reports

# Role of Power BI

### STRENGTHS

- Fast dashboard development
- Interactive analytics
- Microsoft ecosystem integration

### TECHNICAL

- Import Mode (in-memory)
- DirectQuery (live data)
- DAX (calculation layer)

Power BI = Visualization Layer

# The Calyr / Nexus Analogy

## THE PARALLEL

- Calyr: scientific data platform built on Nexus
- BOKU: administrative data platform built on Power BI
- **Same principle** — typed objects, declared contracts, governed outputs

## NEXUS CONTROL LOOP

- Declare → Build → Test → Score → Publish
- Experiments are **declared**, not improvised
- Outputs are valid only if traceable

Data Warehouse = Nexus for administrative data

# Nexus Layer

## CLASSICAL BI

Tables → Dashboards One domain · one tool · one query Efficient, but limited

## NEXUS APPROACH

Data → Typed Objects → Models

- Time domain (kinetics, SPR)
- Reciprocal space (SAXS)
- Real space (geometry)
- Administrative domain (KPIs, reports)

Reports declared, not improvised → Analytical Platform

# Final Vision

**FROM**

- Static reporting
- Manual ETL processes
- Isolated data silos

**TO**

- Scalable data platform
- AI-ready infrastructure
- Integrated analytics
- Nexus foundation

Data Warehouse = Foundation